	AzureSky		
	Statistical Design	Document no: ASMF 7.2.2	
	Compiled by:	Tshepo Maponya – Laboratory Supervisor	
	Approved by:	Sifiso Nyambi – Laboratory Manager	

Procedure for Statistical Design

This procedure outlines the statistical design for the coal proficiency testing (PT) scheme. It provides detailed specifications in accordance with ISO 17043 and related statistical methods from ISO 13528:2022.

1. Design Elements and Procedure Specifications

Design Element (ISO 17043 Ref.)	Procedure Specification
PT Scheme / Measurands	Analysis of Coal * Inherent Moisture (SANS 5925) * Volatile Matter (ISO 562) * Calorific Value (ISO 1928) * Ash (ISO 1171) * Total Sulphur (ASTM D4239) * Ash Fusion Temperature * Hardness (HGI) * Ash in Analysis (AI)
PT Item & Frequency (7.2.2.3d, 7.2.2.3j)	Item: One bottle (approx. 150g) of pulverized coal. Frequency: Monthly.
Objectives (7.2.2.1)	1. To evaluate the performance of participating laboratories (interlaboratory comparison). 2. To establish robust consensus values for the measurands, suitable for assigning property values.
Data Type (7.2.2.1)	Quantitative. All measurands are reported as numerical results.
Determination of Assigned Value (xA) (7.2.2.3a)	The robust consensus value of participants' results, determined using ISO 13528:2022, Algorithm A.
Determination of σpt (7.2.2.3e)	The robust consensus standard deviation of participants' results, determined using ISO 13528:2022, Algorithm A.
Performance Evaluation (7.2.2.2)	z-scores are used for performance evaluation: z = (xi - xA) / σpt
Minimum Participants (n) (7.2.2.3b)	A minimum of n=10 participants is required for robust statistical analysis.
Alternative Approach (n < 10) (7.2.2.3b)	If n < 10: * xA is set by the PT Provider's reference laboratory. * σpt is set from historical data from previous, larger PT rounds. * This change is clearly

	documented in the final report.
Handling of Outliers (7.2.2.3h)	Outliers are not removed before analysis. The use of Algorithm A ensures robust values are determined. A result is identified as an outlier if its final performance score is $\|z\| \geq 3.0$.
Handling of Test Methods (7.2.2.3f)	All specified participant methods (e.g., SANS, ISO, ASTM) are considered technically equivalent and are combined into a single statistical population.
Participant MU (7.2.2.3g)	Participants are not required to report their measurement uncertainty. Performance evaluation is based solely on z-scores.
Repeats / Reporting (7.2.2.3d, 7.2.2.3c)	Participants report a single determination. Results are analyzed and reported to specified decimal places: 1 decimal (Ash, IM, VM); 2 decimals (CV, TS); 0 decimals (HGI, AI).

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2. Narrative Statistical Design Description

PT Scheme: Analysis of Coal

PT Item (Matrix): One bottle (approx. 150g) of pulverized coal.

Measurands: Inherent Moisture, Volatile Matter, Calorific Value, Ash, Total Sulphur, Ash Fusion Temperature, Hardgrove Grindability Index (HGI), Ash in Analysis (AI).

Data Type: Quantitative.

Frequency (7.2.2.3j): Monthly.

Statistical Model (7.2.2.2): The assigned value (xA) will be the robust consensus value of participants' results. The standard deviation for proficiency assessment (σpt) will be the robust consensus standard deviation of participants' results, both determined using ISO 13528:2022, Algorithm A.

Performance Evaluation: Participant performance is evaluated using z-scores, calculated as $z = (xi - xA) / \sigma pt$. Interpretation: $|z| \leq 2.0$ – Acceptable; $2.0 < |z| < 3.0$ – Questionable; $|z| \geq 3.0$ – Unacceptable.

Minimum Participants (7.2.2.3b): A minimum of 10 participants is required for statistically meaningful results.

Alternative Approach ($n < 10$): For fewer than 10 participants, xA is determined by the PT Provider's reference laboratory and σpt from historical data. A z-score-based evaluation remains, with this adjustment documented in the final report.

Test Methods (7.2.2.3f): All SANS, ISO, and ASTM methods are considered equivalent and pooled for analysis.

Outliers (7.2.2.3h): Algorithm A automatically accounts for outliers. A result with $|z| \geq 3.0$ is identified as an outlier.

Measurement Uncertainty (7.2.2.3g): Reporting MU is not required; evaluation uses z-scores only.

Reporting (7.2.2.3c): Decimal precision – Ash, IM, VM: 1 decimal; CV, TS: 2 decimals; HGI, AI: 0 decimals.